1.3 Time Complexity of Multi-head Self-Attention

In a transformer, the multi-head self-attention mechanism processes an input sequence of length T and embedding dimension D using H attention heads. Each attention block performs three major computational steps:

1) **Linear projections for Q, K, and V:**
Each token embedding (of size D) is projected into Query (Q), Key (K), and Value (V) matrices through linear transformations with weight matrices of shape $D \times D$. The total computation cost for these projections is
$$O(3TD^2) \approx O(TD^2).$$

2) **Scaled dot-product attention for each head:**
Each head has dimensionality D/H. The attention score matrix is computed as $QK^\top$, which requires
$$O(T^2D/H)$$
operations. The subsequent softmax and multiplication with V are of the same order. Since there are H heads, the total complexity for all heads becomes
$$O(HT^2D/H) = O(T^2D).$$

3) **Output projection:**
The concatenated outputs of all heads (size $T \times D$) are passed through a final linear layer with weight matrix $D \times D$, requiring
$$O(TD^2).$$

Combining all parts, the total time complexity of the multi-head attention block is
$$\boxed{O(T^2D + TD^2)}.$$

In practice, when the sequence length T is much larger than the embedding dimension D, the quadratic term $O(T^2D)$ dominates. Therefore, the effective time complexity of multi-head self-attention is
$$\boxed{O(T^2D)}.$$

The number of heads H does not change the overall asymptotic complexity, because the per-head dimension D/H compensates for the increased number of heads.